Data & Knowledge Graph Engineer

AI / Higher Education POC

Full-Stack AI + Education Expert | Seiichi Tanaka

5AI Orchestration | 15+ MCP Integrations | 30yr STEM Teaching | Neo4j + GraphRAG | $55/hr

$322.7B	Only 27	2.75x	65%+
AI in Ed Market 2030	Neo4j Freelancers on Upwork	KG+LLM Accuracy Boost	EdTech Failure Rate

Why Seiichi Tanaka?

The only candidate who is both educator AND engineer

EDU 30 Years of STEM Education

Deep understanding of how students learn, cognitive load theory, and curriculum design. Educational knowledge graphs require domain-specific entity relationships that pure engineers cannot model correctly.

5AI 5AI Orchestration System

Unified Claude + Gemini + ChatGPT + Grok + Manus workflow. Research shows multi-agent LLM systems achieve 140x accuracy improvement over single-agent approaches (arXiv 2511.15755).

MCP 15+ MCP Server Integrations

Production-grade Model Context Protocol ecosystem. Neo4j MCP server enables AI agents to interact with knowledge graphs via Cypher queries. Standardized protocol for connecting all education data sources.

COG Cognitive Overlay Layer Design

Personalized learning path recommendation through student cognitive state mapping. Graph-based Cognitive Diagnosis Models track student-skill-question interactions on heterogeneous cognitive graphs.

System Architecture

Cognitive Overlay Layer: Student knowledge state tracking + Personalized learning path recommendation

Confidential Proposal

Development Plan & Delivery

01 DISCOVER
Schema Analysis & Data Mapping

02 MODEL
Knowledge Graph Ontology Design

03 BUILD
Neo4j + GraphRAG Core Pipeline

04 INTEGRATE
PeopleSoft + Canvas Event Streams

05 DELIVER
POC Demo + Docs & Handoff

Milestone Delivery Plan

Milestone	Deliverables	Timeline	Rate
M1: Schema & Graph Model	PeopleSoft/Canvas schema analysis, Neo4j ontology design, data pipeline prototype	Week 1-2	$55/hr
M2: GraphRAG Pipeline	Knowledge graph population, GraphRAG indexing, vector store integration, Cypher queries	Week 2-3	$55/hr
M3: Inference Layer	LLM inference layer, cognitive overlay prototype, student success prediction model	Week 3-4	$55/hr
M4: Integration & Demo	Event-driven PeopleSoft/Canvas sync, end-to-end testing, documentation, POC demo	Week 4-6	$55/hr

Key Features Included

GraphRAG
Microsoft's graph-enhanced retrieval system

Real-time Sync
Event-driven PeopleSoft to Canvas pipeline

Neo4j KG
Production knowledge graph with Cypher

LLM Inference
Multi-model AI reasoning layer

Vector Store
Embedding-based semantic search

MCP Protocol
Standardized AI-data connection layer

Proven Track Record

Project	Technology	Result
5AI Orchestration System	Claude, Gemini, GPT, Grok, Manus	Multi-AI unified workflow
Bybit Trading Platform	Python, API, WebSocket	Live trading automation
MCP Server Integrations	15+ servers, TypeScript, Python	Production MCP ecosystem
STEM Education Platform	React, Vite, PWA (20 apps/week)	K-12 to university level
Pine Script Trading Suite	TradingView, Smart Money Concepts	Multi-instrument analysis

$55/hr
HOURLY RATE

4-6 Weeks
TIMELINE

ASAP
READY TO START

Let's build an AI-first education platform that actually understands how students learn.